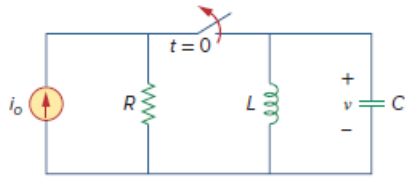


Problem

Find $v(t)$ for $t > 0$ in the circuit in Fig. 16.68.

Figure 16.68:



Step-by-step solution

Step 1 of 5

Refer to figure 16.68 for the circuit in the textbook.

For $t < 0$, circuit is in steady state.

In steady state inductor acts as a short circuit and capacitor acts as an open circuit.

Redraw the circuit for $t < 0$ as shown in figure 1.

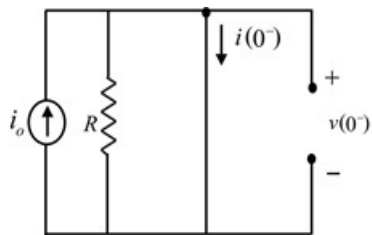


Figure 1

Step 2 of 5

Calculate the initial value of inductor current.

$$i(0^-) = i_o$$

The inductor doesn't allow sudden changes in current.

$$i(0^+) = i(0^-) = i(0) = i_o$$

Calculate the initial value of capacitor voltage.

$$v(0^-) = 0 \text{ V}$$

The capacitor doesn't allow sudden changes in voltage.

$$v(0^+) = v(0^-) = v(0) = 0 \text{ V}$$

Step 3 of 5

Redraw the circuit for $t > 0$ as shown in figure 2.

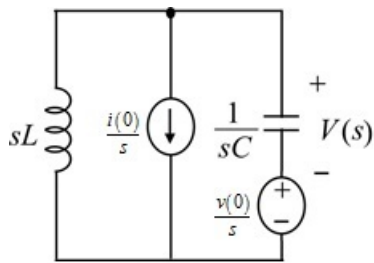


Figure 2

Step 4 of 5

Apply Kirchhoff's current law at the top node of the circuit shown in figure 2.

$$\frac{V(s)}{sL} + \frac{i(0)}{s} + \frac{V(s) - \frac{v(0)}{s}}{\frac{1}{sC}} = 0$$

Substitute i_o for $i(0)$, and 0 for $v(0)$.

$$\frac{V(s)}{sL} + \frac{i_o}{s} + \frac{V(s)}{\frac{1}{sC}} = 0$$

$$V(s) \left(\frac{1}{sL} + sC \right) = \frac{-i_o}{s}$$

$$V(s) \left(\frac{1 + s^2 LC}{sL} \right) = \frac{-i_o}{s}$$

Simplify further,

$$\begin{aligned} V(s) &= \frac{-i_o L}{1 + s^2 LC} \\ &= \frac{-i_o L}{LC \left(s^2 + \frac{1}{LC} \right)} \\ &= \frac{-i_o L}{\sqrt{LC}} \left[\frac{1/\sqrt{LC}}{s^2 + \left(1/\sqrt{LC} \right)^2} \right] \end{aligned}$$

Apply the inverse Laplace transform.

$$\begin{aligned} v(t) &= -\frac{i_o L}{\sqrt{LC}} \sin \left(\frac{t}{\sqrt{LC}} \right) u(t) \text{ V} \\ &= -i_o \sqrt{\frac{L}{C}} \sin \left(\frac{t}{\sqrt{LC}} \right) u(t) \text{ V} \\ &= \frac{-i_o}{\sqrt{\frac{C^2}{LC}}} \sin \left(\frac{t}{\sqrt{LC}} \right) u(t) \text{ V} \\ &= \frac{-i_o}{\sqrt{LC}} \sin \left(\frac{t}{\sqrt{LC}} \right) u(t) \text{ V} \end{aligned}$$

Step 5 of 5

The resonant frequency of a LC network is,

$$\omega = \frac{1}{\sqrt{LC}}$$

Therefore, the voltage $v(t)$ is,

$$v(t) = \frac{-i_o}{\omega C} \sin(\omega t) u(t) \text{ V}$$

Rewrite as,

$$v(t) = \frac{i_o}{\omega C} \cos(\omega t + 90^\circ) u(t) \text{ V}$$

Therefore, the voltage $v(t)$ for $t > 0$ is $\boxed{\frac{i_o}{\omega C} \cos(\omega t + 90^\circ) u(t) \text{ V}}$.